

FMUSER FUQ-4654E Encoder Modulator Multi-input SDI-HDMI-YPbPr-CVBS-ASI DVB-S2X RF Output HEVC H.265 Encoding APSK Modulation Low-Latency for Professional Broadcast

Product Overview

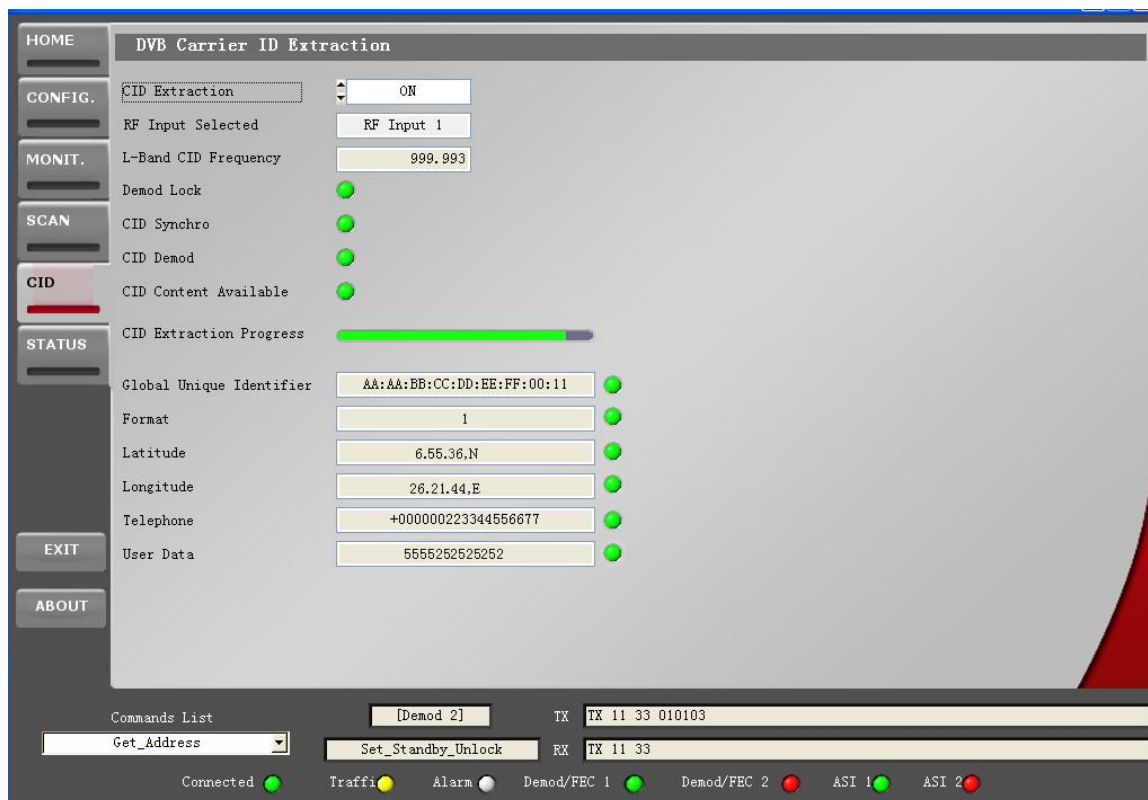


The FUQ-4654E integrates DVB-S/S2/S2X encoding and modulation, featuring pluggable modules for SDI/HDMI/YPbPr/CVBS inputs and ASI input for re-muxing. It encodes SD/HD programs and modulates them to RF output with low-latency modes. Controlled via web-based NMS or front-panel, this encoder modulator suits TV stations and DSNG uplinks for efficient broadcast transmission.

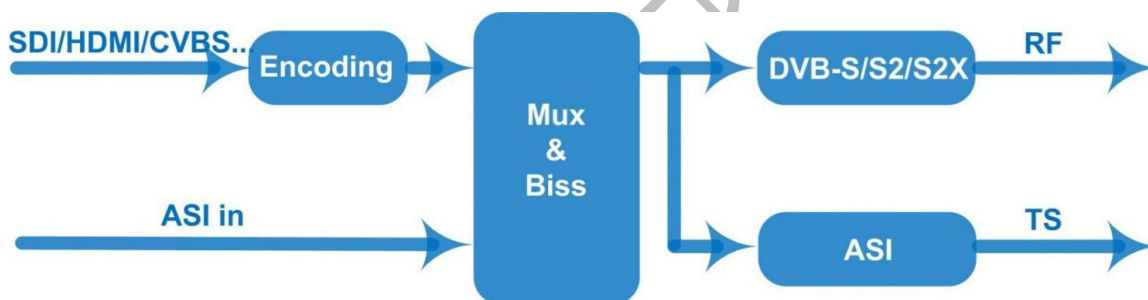
Key feature

- Integrated pluggable modules for selecting HEVC/H.265, MPEG-4 AVC/H.264, or MPEG2 encoding.
- Supports multi-format I/O including SDI/HDMI/YPbPr/CVBS/ASI inputs and DVB-S/S2/S2X RF output (950–2150MHz).
- Features advanced modulation with configurable QPSK/8PSK/8APSK/16APSK/32APSK constellations and BISS encryption.
- Managed via Web NMS or front-panel with low-latency modes and 10MHz reference clock.

CID TEST SAMPLE ILLUSTRATION



Principle Chart



Technical Specifications



Single HDMI/SDI/YPbPr Input

**H.265/H.264 Video Encoding *Up to 4 stereo audio inputs *OSD Insertion*



Video

Encoding	Encoding	HEVC/H.265; MPEG-4 AVC/H.264	HEVC/H.265; MPEG-4 AVC/H.264	HEVC/H.265; MPEG-4 AVC/H.264	HEVC/H.265; MPEG-4 AVC/H.264
Input	Input	HDMI/SDI/YP bPr	HDMI/SDI/YP bPr	HDMI/SDI/YP bPr	HDMI/SDI/YP bPr
Resolution	Input	Input	Input	Output	Interface Applicable
Resolution	1920×1080_60P, 1920×1080_60i, 1920×1080_59.94P, 1920×1080_59.94i	1920×1080_60P, 1920×1080_60i, 1920×1080_59.94P, 1920×1080_59.94i	1920×1080_60P, 1920×1080_60i, 1920×1080_59.94P, 1920×1080_59.94i	1920×1080_60P	. HDMI SDI
Resolution	1920×1080_50P, 1920×1080_50i	1920×1080_50P, 1920×1080_50i	1920×1080_50P, 1920×1080_50i	1920×1080_50P	. HDMI SDI
Resolution	1280×720_60P, 1280×720_59.94P,	1280×720_60P, 1280×720_59.94P,	1280×720_60P, 1280×720_59.94P,	1280×720_60P	. HDMI SDI
Resolution	1280×720_50P	1280×720_50P	1280×720_50P	1280×720_50P	. HDMI SDI

Resolution	1920×1080_60i, 1920×1080_50i, 1280×720_60P, 1280×720_50P	1920×1080_60i, 1920×1080_50i, 1280×720_60P, 1280×720_50P	1920×1080_60i, 1920×1080_50i, 1280×720_60P, 1280×720_50P	1920×1080_60i, 1920×1080_50i, 1280×720_60P, 1280×720_50P	YPbPr
Bit rate	Bit rate	Bit rate	4Mbps~20Mbps (for MPEG-4 AVC/H.264) 0.5Mbps~20Mbps (HEVC/H.265)	4Mbps~20Mbps (for MPEG-4 AVC/H.264) 0.5Mbps~20Mbps (HEVC/H.265)	4Mbps~20Mbps (for MPEG-4 AVC/H.264) 0.5Mbps~20Mbps (HEVC/H.265)
Chroma	Chroma	Chroma	4:2:0	4:2:0	4:2:0
Rate Control	Rate Control	Rate Control	CBR/VBR	CBR/VBR	CBR/VBR
GOP Structure	GOP Structure	GOP Structure	IBBP, IPPP	IBBP, IPPP	IBBP, IPPP
Advanced Pretreatment	Advanced Pretreatment	Advanced Pretreatment	De-interlacing, Noise Reduction, Sharpening	De-interlacing, Noise Reduction, Sharpening	De-interlacing, Noise Reduction, Sharpening

Audio

encoding	MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough
Input	Digital: HDMI/SDI/AES; Analog: XLR
Channel	AES/SDI: Up to 4 stereos HDMI: 1 stereo XLR: Up to 2 stereos
Sample rate	48KHz
Bit rate	48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)
Audio Gain	0-255



Single SDI Input

Video

Encoding	MPEG2; MPEG4 AVC/H.264
Input	SDI*1
Resolution	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P 720*480_60i, 720*576_50i

Audio

encoding	MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC
Sample rate	48KHz
Bit rate	64kbps, 96kbps, 128kbps, 192kbps, 256kbps, 320kbps



Single HDMI Input

Video

Encoding	MPEG2; MPEG4 AVC/H.264
Input	HDMI*1
Resolution	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P 720*480_60i, 720*576_50i

Audio

encoding	MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC
Sample rate	48KHz
Bit rate	64kbps, 96kbps, 128kbps, 192kbps, 256kbps, 320kbps



HDMI/YPbPr/CVBS 3-in-1 Input

Video (HDMI)	Video (HDMI)
Encoding	MPEG2; MPEG4 AVC/H.264
Interface	HDMI*1
Resolution	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P
Audio (HDMI)	Audio (HDMI)
Encoding	MPEG1 Layer II; MPEG2-AAC; MPEG4-AAC
Interface	HDMI*1
Sample rate	48KHz
Bit rate	64/96/128/ 192/256/320kbps
Video (YPbPr/CVBS)	Video (YPbPr/CVBS)
Encoding	MPEG2; MPEG4 AVC/H.264
Interface	YPbPr*1/CVBS *1
Resolution	CVBS: 720x576_50i (PAL); 720x480_60i (NTSC) YPbPr: 1920*1080_60i, 1920*1080_50i; 1280*720_60p, 1280*720_50P
Audio (L/R)	Audio (L/R)
Encoding	MPEG1 Layer II; MPEG2-AAC; MPEG4-AAC
Interface	1*Stereo/2*mono
Sample rate	48KHz
Bit rate	64/96/128/ 192/256/320kbps



2*(S-Video + YPbPr + CVBS) Inputs

Video

Encoding	Option 1: MPEG-2 MP@ML(4:2:0) Option 2: MPEG-2 & MPEG-4 AVC/H.264 (4:2:0)
Input	2*S-Video/YPbPr/CVBS (RCA Type)
Resolution	720*576 (PAL); 720*480 (NTSC)
Bit rate	Option 1: 1~19.50 Mbps; Option 2: 1~10.00 Mbps

Audio

encoding	Option 1: MPEG1 Layer II Option 2: MPEG1 Layer II; LC-AAC; HE-AAC and Dolby Digital AC3 2.0(Optional)
Input	2 Stereo/Mono
Sample rate	48KHz
Bit rate	64kbps, 96kbps, 128kbps, 192kbps, 256kbps, 320kbps

Generals

Input Interface	2*CVBS; 1*SDI; 1*HDMI; 1*(HDMI/SDI/YPbPr 3-in-1); 1*(HDMI/YPbPr/CVBS 3-in-1) input optional ASI input	2*CVBS; 1*SDI; 1*HDMI; 1*(HDMI/SDI/YPbPr 3-in-1); 1*(HDMI/YPbPr/CVBS 3-in-1) input optional ASI input	2*CVBS; 1*SDI; 1*HDMI; 1*(HDMI/SDI/YPbPr 3-in-1); 1*(HDMI/YPbPr/CVBS 3-in-1) input optional ASI input	2*CVBS; 1*SDI; 1*HDMI; 1*(HDMI/SDI/YPbPr 3-in-1); 1*(HDMI/YPbPr/CVBS 3-in-1) input optional ASI input
Output	DVB-S/S2/S2X RF output ; ASI output as mirror of RF	DVB-S/S2/S2X RF output ; ASI output as mirror of RF	DVB-S/S2/S2X RF output ; ASI output as mirror of RF	DVB-S/S2/S2X RF output ; ASI output as mirror of RF
Modulation Section	Standard	DVB-S	DVB-S2	DVB-S2X
Modulation Section	Outer coding	RS Coding	BCH Coding	BCH Coding
Modulation Section	Inner coding	Convolution	LDPC Coding	LDPC Coding

Modulation Section	Constellation	QPSK	QPSK, 8PSK, 16APSK, 32APSK	QPSK, 8PSK, 8APSK, 16APSK, 32APSK
Modulation Section	FEC/ Convolution Rate	1/2, 2/3, 3/4, 5/6, 7/8	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10, 2/3-L, 32/45, 11/15, 7/9
Modulation Section	Roll-off Factor	0.2, 0.25, 0.35	0.2, 0.25, 0.35	0.05, 0.10, 0.15, 0.2, 0.25, 0.35
Modulation Section	Symbol Rate	0.5~45Msps	0.5~40Msps (32APSK); 0.5~45Msps (QPSK/8PSK/16APSK)	0.5~40Msps(32APSK); 0.5~45Msps (QPSK/8PSK/16APSK)
Modulation Section	RF Range	950-2150MHz	950-2150MHz	950-2150MHz
Modulation Section	Output level	+3dBm~-28.5dBm	+3dBm~-28.5dBm	+3dBm~-28.5dBm
BISS	Mode 1, Mode E	Mode 1, Mode E	Mode 1, Mode E	Mode 1, Mode E
System	Local Control	LCD + control buttons	LCD + control buttons	LCD + control buttons
System	Remote Control	Web NMS	Web NMS	Web NMS
System	Low Latency Mode	Normal, Mode 1, Mode 2	Normal, Mode 1, Mode 2	Normal, Mode 1, Mode 2

System	Language	English	English	English
Physical Specification	Demission	410×460×44mm (W*L*H)	410×460×44mm (W*L*H)	410×460×44mm (W*L*H)
Physical Specification	Temperature	0~45■(operation), -20~80■(storage)	0~45■(operation), -20~80■(storage)	0~45■(operation), -20~80■(storage)
Physical Specification	Power Supply	AC 100-240V 50/60Hz 2A	AC 100-240V 50/60Hz 2A	AC 100-240V 50/60Hz 2A

Order Guide

	FU-4654A	FU-4654C	FU-4654D
DVB-S/S2	●	●	●
DVB-S2X			●
RF Output(950-2150MHz)		●	●
IF Output(50-200 MHz)	●		

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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